Electronic Counter-Countermeasures (ECCM)

Quick briefing

Electronic counter-countermeasures are taught as Electronic Protection (EP). Use communications discipline and continuity planning to preserve friendly service, then distinguish an observed disruption from its possible causes.

Know the facts

Electronic Protection · EP safeguards friendly capabilities in the electromagnetic environment. The lesson treats it as an all-unit responsibility.

EMCON and radio discipline · Emission control governs what may transmit under given conditions. The lesson names Level 3 AMBER as its field baseline and teaches necessary, brief transmissions.

Encryption and anti-jam measures · Encryption protects information content. Anti-jam measures address the ability to keep communicating despite interference.

Possible causes of disruption · Sudden static or lost service can stem from interference, equipment faults, terrain or distance, or jamming. A symptom alone leaves the cause uncertain.

PACE · Primary, Alternate, Contingency and Emergency. Rehearse when to switch and how to maintain essential exchange on the approved backup.

GNSS disruption · Jamming can deny reliable reception. Spoofing can make a receiver show misleading position or time. Cross-check inconsistent indications and report their effects.

Understand the reasoning

Keep the essential service going while someone checks the evidence. Waiting for a confident diagnosis can unnecessarily delay the exercise.

Record the service lost, the circumstances and the actions taken. Clear observations help the recipient assess competing explanations.

Philippine Army classroom exercise

During a Philippine Army convoy tabletop, an instructor injects intermittent static on the primary voice service near an exercise checkpoint. An alternate passes a test message. One crew also reports an equipment warning.

Decision · What should the signal team do first, and what should it report?

Worked analysis · Use the rehearsed trigger and approved alternate to keep the essential traffic moving. Check for local equipment faults and compare symptoms across stations and conditions. Report the observed static, mission effect and actions taken through the exercise channel, with the cause marked as unresolved.

How to practice

1 · Identify the essential traffic and the symptom observed in the instructor's interference drill.

2 · Apply the exercise's rehearsed PACE trigger and confirm that the required message reaches the recipient.

3 · Compare operator observations and check plausible equipment and environmental causes under instructor supervision.

4 · Send a training report covering who, when, where, affected service, symptom, effect and actions. Separate observations from the suspected cause.

Check yourself

Which measure protects message content, encryption or anti-jam measures? · Encryption.

What should a PACE rehearsal confirm in addition to naming the four means? · When to switch and whether the recipient can still receive the essential traffic.

Course source

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